Jashore University of Science and Technology

Bachelor of Science Electrical and Electronic Engineering

1st semester of 1st year, Session: 2025 – 2026

Course no.: 053311 PHY 1101 **Course title: Physics**

Assignment no.: 01 **Date: 10 October 2026**

Deadline for submission: 30 October 2026, 10:00 AM

1. What is interference of light? Under what conditions interference of light take place?

2. In case of Young's double slit experiment, show that $I = 4I_0 \cos^2 \frac{\delta}{2}$, where the symbols carry usual meaning.

3. Define diffraction of light. What is Fresnel and Fraunhofer class of diffraction?

4. For single slit diffraction show that $I = I_0 \left(\frac{\sin\beta}{\beta}\right)^2$, where the symbols carry usual meaning.

5. Define polarization of light. How can elliptically and circularly polarized light be formed from plane polarized light?

6. What are the thermodynamic functions? Derive the Maxwells relations related to the thermodynamic functions H and G.

7. Define efficiency of an engine. State and prove the Carnot's theorem.